When Does Compressed KV Start Lying?

Token-Level Drift in KV Cache Compression

ExactKV Project

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If you read five minutes. ExactKV compares compressed-KV greedy decoding to full-precision greedy decoding. *Drift %* = share of prompts with ≥1 differing token; *first divergence* = that token index. (1) matched hierarchy: `int4_sim` Mistral **0%/23%/97%**, Llama **17%/26%/91%** (code/tools/reading) at shared budgets; (2) BFCL length still scales 9%→62%; (3) unguarded lossy can ship wrong fields; ExactKV recovers full-KV-valid outputs. Details: Appendix A / site / `REPRODUCE.md`.

Abstract

Lossy KV-cache compression is widely deployed, but most evaluations report aggregate quality and stop there. ExactKV is a compressor-agnostic **crash-test framework** that measures first-divergence index, draft acceptance, unguarded lossy damage, and `exactkv_failure` under verifier-mediated semantics.

Across **8,132 GPU cells** on Llama-3.1-8B and Mistral-7B (Appendix benchmark card)[1], four main findings (glance table below):

1. **Matched budgets:** Mistral 0%/23%/97%, Llama 17%/26%/91% (`int4_sim`); H2O eviction 100% on LongBench.
2. **Within-task length:** BFCL `int4_sim` 9%→62% (mnt 16→256).
3. **Three failure modes:** near-tie (int8), distribution shift (`int4_sim`), attention destruction (H2O) over 1,103 divergent cells.
4. **Recovery:** BFCL validity **318/318** when full-KV valid; MBPP 12/12 (Llama); LongBench overlap 100% vs full-KV.

`int6_sim` and `int4_per_vec_sim` are GPU-validated non-catastrophic on structured tasks (extended validation, 1,568 cells). **Faithful upstream adapters** (1,568 appendix cells): TurboQuant is task-conditional (∼65% LongBench); `int8` remains the only non-catastrophic real compressor in that grid (8.3% combined). Drift is jointly governed by task, length, compressor class, and granularity. Results are **not** benchmark scores, production claims, or a reproduction of VeriCache [8].

[1] Six built-in compressor classes: `noop`, `int8`, `int6_sim`, `int4_per_vec_sim`, `int4_sim`, H2O-style eviction (`h2o_sim`). External adapters evaluated separately (Section 14.1). `exactkv_failures=0` is a harness invariant, not the scientific headline.

1 Introduction

Lossy KV-cache compression is widely deployed as a transparent optimization for LLM inference. Under greedy decoding, a single perturbed logit can flip the argmax, after which trajectories diverge. Aggregate metrics (perplexity, LongBench, task accuracy) average over this divergence and hide *where* it begins, *how severe* it is, and *which compressor design choices* drive it.

ExactKV reframes KV-cache evaluation around a single diagnostic question: **at what generated token does the compressed-cache path first stop matching full-KV greedy decoding?** We answer this with a compressor-agnostic crash-test framework that runs a draft/verify/commit loop and records first-divergence index, draft acceptance, verifier agreement, and exactness failures per cell.

Contributions:

1. A **compressor-agnostic crash-test framework** with leaderboard-style reporting for token-level drift, first divergence, acceptance, and exactness failures (§3–5).
2. **Empirical map of when compressors drift** — matched task×budget hierarchy, within-task length scaling, eviction vs quantization (glance table; Table 25, §6.12).
3. A **logit autopsy** over 1,103 divergent cells with three failure modes (§10, §15).
4. **Unguarded lossy damage vs verifier recovery** across three families; cost sketch §9.1 (§11).
5. **8,132 headline GPU cells** plus a 1,568-cell faithful-adapter appendix and GPU validation of `int6_sim`/`int4_per_vec_sim` (§6, Sections 13–14).

1.1 Shared problem framing with VeriCache

VeriCache [8] and ExactKV share the observation that lossy KV may look acceptable on short or aggregate metrics but **diverges more as decoding continues**, and catastrophic for code and tool-calling. VeriCache uses compressed KV to **draft** and full KV to **verify/correct**, guaranteeing **identical greedy output** to full-KV inference.

1.2 Where ExactKV diverges from VeriCache

VeriCache is a **serving/system** paper. Its contribution is not merely "draft then verify." It makes that loop practical by: (1) keeping compressed KV on GPU, (2) keeping full KV in CPU/storage, loading only for verification, (3) **cross-resource staggering** (HBM-bound draft vs interconnect-bound verify), (4) reporting serving throughput (up to ∼4× vs full-KV in their evaluation) with identical outputs.

ExactKV is an **evaluation crash-test**. It measures first divergence, acceptance, and exactness failures, *not* deployment throughput. It does *not* reproduce VeriCache scheduling or memory tiering.

1.3 Main Findings at a Glance

#	Finding	Key number
1	Matched task-family hierarchy (both models)	`int4_sim`: Mistral 0%/23%/97%; Llama 17%/26%/91%
2	Generation length (within-task control)	`int4_sim` mnt=16: 9% → mnt=256: 62% (7×)
3	Eviction > quantization	H2O-style 75% kept → 100% LongBench divergence
4	Three distinct failure modes	int8 near-tie (rank 2.4, fdi=22); `int4_sim` shift (rank 3.5, fdi=8); H2O destruction (rank 6.7, fdi=1)
5	Unguarded lossy + recovery	BFCL 318/318; MBPP 12/12 (Llama); LongBench overlap 100% (§9.1)
6	Compressor design curve	int8 → int6 → int4 per-vec → int4 sim → H2O (Table 29)
7	Faithful adapters reinforce hierarchy	TurboQuant ∼0–5% code/tool, ∼65% LongBench; int8 8.3% combined

Table 1: Main findings summary. Cross-family spans are observational; BFCL length is the controlled within-task axis. `exactkv_failures=0` is a harness invariant, not the headline.

2 Definitions

2.1 Reference paths

- **Full-KV reference:** greedy generation with uncompressed KV.
- **Lossy path:** compressed-KV greedy without verifier.
- **ExactKV path:** verifier-mediated draft/verify/commit (`ExactKVGenerator`).

2.2 Metrics (formal)

Table 2: Core metrics (formal definitions).

Metric	Definition
`first_divergence_index`	First position where the *lossy* token ≠ full-KV greedy (before correction).
`acceptance_rate`	`total_accepted / total_drafted` across verification rounds.
`exactkv_failure`	Final ExactKV output ≠ full-KV greedy (verifier/commit failure).
`divergence_rate`	Fraction of cells with lossy-path divergence.
`divergence_stability`	1 − `divergence_rate` (complement).
`compression_ratio`	Stored tensor byte ratio only (not active GPU memory).

Corrections vs rejections: `total_rejected` counts rejected *draft tokens*, `total_corrections` counts verification *rounds* with a correction. For `int4_sim`: 1284/1502 accepted, 218 rejected tokens, **116 correction rounds**.

2.3 Algorithm

```
Prefill -> FullKVState + CompressedKVState
while generated < max_new_tokens:
  DRAFT from compressed KV (up to draft_len tokens)
  VERIFY each token vs full-KV greedy
```

```
  COMMIT accepted prefix, correct on mismatch
  ALIGN compressed state from authoritative full KV
return ExactKV output_ids
```

Greedy decoding only in this release.

3 Method

Each **cell** is (model, compressor, prompt, `max_new_tokens`) with `draft_len=4`, `seed=0`. Three modes per cell: **full**, **lossy**, **exactkv**.
Leaderboard-style score (not an official benchmark ranking):

$$\text{score} = 0.35\,a + 0.25\,v + 0.20\,\hat{d} + 0.10\,(1 - f) + 0.10\,s$$

Symbol	Meaning
a	Mean draft acceptance rate (`total_accepted`/`total_drafted`)
v	Mean verifier agreement ($= a$ on the scale panel in this release)
$\hat{d}$	Normalized first-divergence index $\in [0, 1]$: mean FDI/`max_new_tokens`, or 1.0 if none (later / no divergence is better)
f	ExactKV failure rate (ExactKV output $\neq$ full-KV)
s	Stability subscore (Exp-116 when available; else 1− divergence rate)

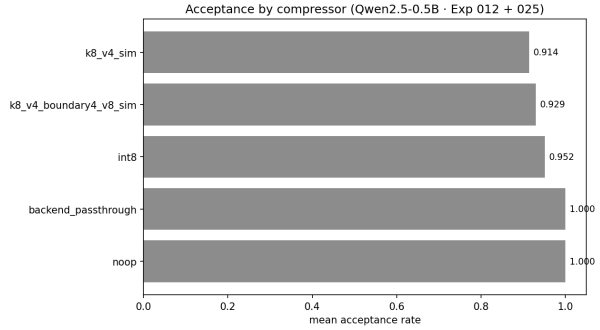
4 Experimental Setup

Table 3: Headline release panel (1,500 cells; Appendix inventory).

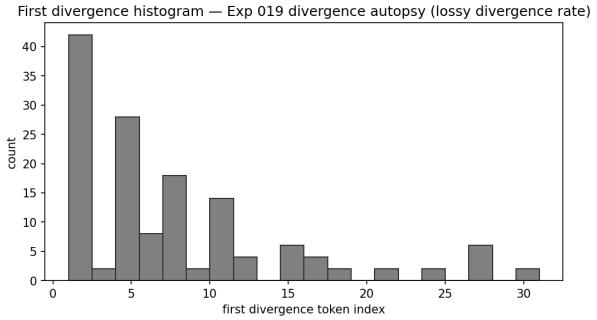
Parameter	Value
Panel ID	`phase_a_unified_panel`
Total cells	**1500** $= 2 \times 5 \times 50 \times 3$
Models	Llama-3.1-8B (750), Mistral-7B (750)
Stack	torch 2.8.0+cu128, transformers 5.12.1, ExactKV (this report)
`draft_len`	4 `max_new_tokens` $\in \{4, 8, 16\}$
Decoding	Greedy only
Prompts	50 variants from 4 stress templates (capital, math, JSON, code)
Categories	capital_france 390, simple_math 390, json_tool 360, code_fn 360
Execution	Sequential per model, `deterministic_mode=false`

Validated compressors: `noop`, `int8`, `int4_sim`. Real KIVI/KVQuant/SnapKV/TurboQuant are not in the headline panel. Unvalidated proxy slots in raw artifacts are excluded from analysis (Limitations §22).

1,560 GPU cells total: 216 Llama-3.1-8B cells across bundled LongBench/RULER/BFCL/HumanEval pilots; 144 MBPP cells across both models; 1,200 BFCL export-50 tool-call drift cells across both models; `exactkv_failures=0` throughout. Not official benchmark scores.



(a) Acceptance by compressor.



(b) First-divergence distribution.

Figure 1: Release benchmark visuals from the headline panel.

Compressor	Cells	Accept.	Div.†	Stab.	FDI§	Fail.
`noop`	300	1.000	0.00	1.00	n/a	0.00
`int8`	300	1.000	0.00	1.00	n/a	0.00
`int4_sim`	300	0.851	0.52	0.48	2.0	0.00

Table 4: Built-in compressors from the headline panel. †Lossy-path divergence fraction. §Mean first-divergence index over divergent cells only. Histogram for `int4_sim`: index 1→78, 3→39, 5→24, 8→13 cells.

5 Results: Validated Compressors

Table 4: built-in compressors with direct ExactKV integration (`noop`, `int8`, `int4_sim`). Aggregates from the headline release panel (300 cells each: 150 per model).

`int8`/`noop`: no lossy divergence. `int4_sim`: drifts in 52% of cells yet `exactkv_failure=0` because the verifier corrects.

Leaderboard-style scores (formula above; Appendix inventory):

Compressor	Model	Score	Accept.	Div. rate†	Stab.	Cells
`noop`	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
`int8`	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
`noop`	Mistral-7B	1.000	1.000	0.000	1.000	150
`int8`	Mistral-7B	0.983	1.000	0.000	0.827	150
`int4_sim`	Llama-3.1-8B	0.859	0.852	0.520	0.480	150
`int4_sim`	Mistral-7B	0.851	0.837	0.507	0.493	150

Table 5: Leaderboard-style scores for validated built-in compressors. †Div. rate = raw lossy-path token divergence fraction (same metric as Table 4). `int8` Mistral: divergence_rate=0.000 (zero cells diverge). `Stab.`=0.827 is the **leaderboard stability subscore**, which is distinct from the formal metric divergence_stability = 1 − divergence_rate (Table 2). Under the formal metric, `int8` Mistral divergence_stability=1.000.

5.1 Metric interpretation

Divergence rate and acceptance rate measure *different* failure modes. Low lossy-path divergence does not imply high verifier acceptance (and vice versa). Example: `int4_sim` at `max_new_tokens=16` can show first divergence at index 8 with acceptance 0.94, how late drift occurs matters.

5.2 Evidence-plus long-context panel

Source: evidence-plus panel (RunPod A5000, June 2026; Appendix inventory). 144 cells = $2 \times 6 \times 2 \times 3 \times 2$, prefill buckets 512/1024, `max_new_tokens` $\in \{16, 32\}$, built-in compressors only. By

Compressor	Cells	Accept.	Div. rate	Fail.
`noop`	48	1.000	0.00	0.00
`int8`	48	1.000	0.00	0.00
`int4_sim`	48	0.985	0.31	0.00

Table 6: Evidence-plus aggregates. `int4_sim`: Mistral 41.7%, Llama 20.8% divergence (24 cells each).

context bucket (all compressors): 512→11.1% divergence, 1024→9.7%. Diagnostic timing: mean 3854 ms, p90 5178 ms per cell (not serving throughput).

5.3 External benchmark smoke panels

Source: Llama-only external smoke (216 cells) and MBPP smoke (144 cells). `exactkv_failures=0` on all panels. 216-cell subset: `noop`/`int8`: 0%. 15 divergent cells (`int4_sim` only).

Panel	**Src**	**Model**	**Ctx**	**MNT**	**n**	**Div.**	**Acc.**	**Fail**	**P90**
LongBench	pilot	Llama	2,4	16,32	72	0.083	0.994	0	8781
RULER 2K–4K	pilot	Llama	2,4	16,32	48	0.083	0.993	0	8793
RULER 8K	pilot	Llama	8	16,32	24	0.083	0.993	0	13640
BFCL	pilot	Llama	1,2	16,32	48	0.063	0.999	0	6342
HumanEval	pilot	Llama	1,2	32	24	0.000	1.000	0	6392

Table 7: Llama-3.1-8B external smoke panels (216 cells, `noop`/`int8`/`int4_sim`). Drift metrics; not official scores. P90 is wall-clock ms.

Panel	**Src**	**Models**	**Ctx**	**MNT**	**n**	**Div.**	**Acc.**	**Fail**	**P90**
MBPP	pilot	Both	0.5,1	16,32	144	0.021	0.999	0	5121

Table 8: MBPP code-drift smoke (144 cells, `noop`/`int8`/`int4_sim`). Not pass@1; no test execution.

Compressor	**Model**	**Cells**	**Div. rate**	**CI_{95} low**	**CI_{95} high**
`noop`	Llama-3.1-8B	200	0.000	0.000	0.019
`noop`	Mistral-7B	200	0.000	0.000	0.019
`int8`	Llama-3.1-8B	200	0.000	0.000	0.019
`int8`	Mistral-7B	200	0.000	0.000	0.019
`int4_sim`	Llama-3.1-8B	200	0.055	0.031	0.096
`int4_sim`	Mistral-7B	200	0.170	0.124	0.228

Table 9: BFCL export-50 tool-call drift panel (1,200 cells, both models, 50 prompts). `exactkv_failures=0`. Source: `bfcl_export_50_raw.json`. Drift measurement only (not JSON-completeness). `max_new_tokens`$\in \{16, 32\}$. Mistral-7B `int4_sim` divergence (17.0%) is 3× Llama-3.1-8B (5.5%).

Notable: LongBench `int4_sim` 25%. RULER 8192 `niah_single` 33%. BFCL pilot `int4_sim` 18.75%. BFCL export-50 `int4_sim` 11.3% (Llama 5.5%, Mistral 17.0%). HumanEval zero drift. MBPP 3/144 divergent. `int8`/`noop`: 0% divergence on all panels.

Overall acceptance (initial 216 ext. pilot cells): Full-acceptance rate 93.7%, Wilson 95% CI [90.5%, 96.8%]. Source: `analysis_pack.json` → `totals.acceptance_full_rate_ci95`.

Panel	n	Div.	Rate	CI_{95} low	CI_{95} high
LongBench pilot	72	6	8.3%	3.9%	17.0%
RULER 2K–4K	48	4	8.3%	3.3%	19.7%
RULER 8K	24	2	8.3%	2.3%	25.8%
BFCL pilot	48	3	6.3%	2.2%	16.8%
HumanEval pilot	24	0	0.0%	0.0%	14.2%
MBPP pilot	144	3	2.1%	0.7%	6.0%
BFCL exp-50 `int4_sim` Llama	200	11	5.5%	3.1%	9.6%
BFCL exp-50 `int4_sim` Mistral	200	34	17.0%	12.4%	22.8%
BFCL exp-50 `int8/noop`	800	0	0.0%	0.0%	0.5%
All 216 ext. cells	216	15	6.9%	4.3%	11.1%

Table 10: Wilson 95% CIs on divergence rate (panel-level). "All 216 ext. cells" refers to the **initial** LongBench/RULER/BFCL/HumanEval pilot panel (first workflow, Llama-only). The full 1,560-cell external smoke total includes MBPP and BFCL export-50.

5.4 KIVI offline compressor panel, real HF results (640 cells)

Source: `kivi_longbench_hf_raw.json` (320 cells) + `kivi_mbpp_hf_raw.json` (320 cells). RunPod RTX A5000, June 2026. `exactkv_failures=0` on all 640 cells.

Compressor	LB div.	LB acc.	MBPP div.	MBPP acc.	Fail.
`noop`	0.0%	1.000	0.0%	1.000	0
`int8`	13.8%	0.994	0.0%	1.000	0
`int4_sim`	91.2%	0.818	5.0%	0.995	0
`kivi_offline`	**100.0%**	**0.004**	**100.0%**	**0.001**	**0**

Table 11: KIVI offline compressor panel (80 cells each × 4 compressors × 2 datasets). `kivi_offline` produced catastrophic **adapter-level** drift in this ExactKV integration (near-zero acceptance). Because this is an offline simulate-path adapter rather than KIVI production CUDA/Triton serving, these results diagnose the adapter/integration path, *not* the KIVI algorithm as deployed. ExactKV detected and corrected all cells (`exactkv_failures=0`). Real HF `int4_sim` drift (LB: 91.2%) greatly exceeds bundled pilot (20.8%).

6 ExactKV-HF-LongBench Drift Panel (v2.6, Complete)

Status: Complete. 720 cells, both models, `exactkv_failures=0`.

Key finding (task-type sensitivity): Real HF LongBench open-text reading/summarization at 2K–8K prefill produces **very high int4_sim divergence** (90.4%), the highest seen across all ExactKV panels. Even int8 shows 24.6% divergence on LongBench tasks, unlike its 0% on BFCL, MBPP, and RULER. Divergence occurs very early (median token 4–6 out of 32–64 generated). `exactkv_failures=0` across all 720 cells.

Table 13 shows per-subset int4_sim divergence. `lcc` (code completion) is notably lower for Mistral (42%), consistent with code tasks being more stable under int4_sim across panels.

Model	Compressor	n	Div.	Rate	CI_{95}	Mean acc.	EKV fail
Llama-3.1-8B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Llama-3.1-8B	int8	120	33	27.5%	[20.3%, 36.1%]	0.988	0
Llama-3.1-8B	int4_sim	120	110	**91.7%**	[85.3%, 95.4%]	0.825	0
Mistral-7B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Mistral-7B	int8	120	26	21.7%	[15.2%, 29.9%]	0.988	0
Mistral-7B	int4_sim	120	107	**89.2%**	[82.3%, 93.6%]	0.861	0
All (int4_sim)	all	**240**	**217**	**90.4%**	**[86.1%, 93.5%]**	0.843	**0**

Table 12: ExactKV-HF-LongBench v2.6 (720 cells, 10 real HF subsets, contexts 2K/4K/8K, `mnt`=32/64, both models). int4_sim: **90.4%** divergence on long-context open-text tasks, highest across all ExactKV panels. int8 shows 24.6%, unlike its 0% on BFCL/MBPP/RULER. `exactkv_failures=0`. Median first-div index: 4 (Llama) / 6 (Mistral).

Subset	Type	Llama int4_sim	Mistral int4_sim
2wikimqa	multi-hop QA	12/12 (100%)	12/12 (100%)
gov_report	summarization	11/12 (92%)	12/12 (100%)
hotpotqa	multi-hop QA	12/12 (100%)	12/12 (100%)
lcc	**code completion**	**11/12 (92%)**	**5/12 (42%)**
multifieldqa_en	multi-field QA	6/12 (50%)	10/12 (83%)
narrativeqa	reading comp.	12/12 (100%)	12/12 (100%)
passage_retr.	doc. retrieval	12/12 (100%)	12/12 (100%)
qasper	academic QA	12/12 (100%)	12/12 (100%)
samsum	dialogue summ.	10/12 (83%)	12/12 (100%)
trec	classification	12/12 (100%)	8/12 (67%)

Table 13: Per-subset int4_sim divergence in HF LongBench v2.6 (2K/4K/8K context, `mnt`=32/64). `lcc` (code completion) shows lower divergence for Mistral (42%) vs. reading-comprehension tasks (100%), consistent with code-task stability across all ExactKV panels. `exactkv_failures=0` for all 720 cells.

7 BFCL Tool-Call Validity Panel (v2.7, Complete, 1,200 cells)

The 48-cell BFCL pilot used `max_new_tokens` $\in$ `{16, 32}`, too short for complete JSON tool calls. A dedicated 1,200-cell validity panel (`max_new_tokens=128/256`) was completed with both models.

Compressor	Model	n	Div.	Rate	EKV fail
noop	Llama	200	0	0.0%	0
noop	Mistral	200	0	0.0%	0
int8	Llama	200	0	0.0%	0
int8	Mistral	200	3	1.5%	0
int4_sim	Llama	200	90	**45.0%**	0
int4_sim	Mistral	200	111	**55.5%**	0
All (1,200)	both	1,200	204	**17.0%**	**0**

Table 14: BFCL tool-call validity panel v2.7 (1,200 cells, mnt=128/256, both models). `int4_sim` diverges 45.0% on Llama and 55.5% on Mistral with long generation, vs 11% at mnt=16/32, confirming generation length as a primary driver. `int8` is near-zero. `exactkv_failures=0` throughout. Artifact: `bfcl_validity_v27_merged_raw.json`.

8 H2O-Style Token-Eviction Panel (v2.8, Complete, 800 cells)

Status: Complete. 800 cells, 5 eviction variants, both models, `exactkv_failures=0`. Artifact: `h2o_v28_merged_raw.json`.

Eviction mechanism: `h2o_sim_75`, `h2o_sim`, and `h2o_sim_25` keep approximately 75%, 50%, and 25% of prefill tokens using a simplified heavy-hitter-style eviction heuristic (cumulative attention score across heads). Lowest-scoring tokens are evicted after the full prefill. retained tokens include attention sinks (positions 0–3) plus the highest-attention tokens by recency/score. This approximates the H2O [9] eviction policy but omits the original online streaming update and GPU-efficient sparse-attention kernel. Results characterize the eviction compressor *class*, not the published H2O system.

Key result: Even mild eviction (75% of tokens kept) causes **100% divergence** on LongBench reading tasks for both models. The H2O-style simulation reaches mean lossy rank 6.5–6.7 at divergence (vs. rank 3.5 for `int4_sim`), confirming attention destruction as a qualitatively distinct failure mode. `exactkv_failures=0` across all 800 cells.

9 Cross-Panel Divergence Analysis: Task-Type Sensitivity

Table 15 summarises int4_sim, int8, and noop divergence rates across completed ExactKV panels. Cross-family rates are **observational**.

Threat / confound. Cross-family comparisons (MBPP vs BFCL vs LongBench) change prompts, context buckets, and `max_new_tokens` together. The memorable `int4_sim` span 6%→90% is *not* a pure causal claim that task type alone drives drift. Controlled axes: BFCL mnt 16→256 (fixed tool-call family) and LongBench 2K/4K/8K (fixed reading family). A matched task×context×`max_new` factorial with shared budgets confirms the hierarchy on both models (Table 25); prompts still differ by family.

Panel	Task	Ctx	mnt	Models	noop	int8	int4_sim
Headline (v1)	mixed	∼500	16/32	L+M	0.0%	0.0%	**51.3%**
Evidence-plus (v2)	mixed	512/1024	16	L+M	0.0%	0.0%	**31.2%**
MBPP code-drift	code	512/1024	16/32	L+M	0.0%	0.0%	**6.2%**
BFCL export-50	tool-call	1K–2K	16/32	L+M	0.0%	0.0%	**11.2%**
BFCL validity v2.7	tool-call	1K–2K	128/256	L+M	0.0%	0.8%	**50.3%**
HF LongBench v2.6	reading/summ.	2K–8K	32/64	L+M	0.0%	24.6%	**90.4%**

Table 15: Cross-panel int4_sim/int8/noop divergence rates (observational across families). noop=0% throughout. Memorable span: code 6% < tool short 11% < tool long 50% < reading 90% — confounded with context/`max_new`. Within-task BFCL length 11%→50% as mnt 16→256. Harness invariant `exactkv_failures=0`. L=Llama-3.1-8B, M=Mistral-7B.

10 Mechanistic Analysis: Divergence Autopsy

The cross-panel observational hierarchy is supported by a token-level mechanistic analysis. ExactKV stored top-5 full-KV and lossy logit distributions at each divergence point (`--store-top-k-logits`), enabling a logit autopsy across 1,103 divergent cells from v2.6, v2.7, and v2.8 panels.

Three qualitatively distinct failure modes emerge:

Compressor	n div	top-1 flip	near-tie	mean margin	mean rank	med. fdi
int8	62	69%	**66%**	0.116	2.4	22
int4_sim	563	83%	26%	0.305	3.5	8
h2o_sim_75	160	**100%**	0%	0.545	6.5	1
h2o_sim	160	**100%**	0%	0.536	6.7	1
h2o_sim_25	158	**100%**	1%	0.551	6.5	1

Table 16: Logit autopsy across 1,103 divergent cells. near-tie: full-KV margin < 0.05. fdi = first-divergence index. Three distinct failure modes: (1) *near-tie noise* (int8), (2) *distribution shift* (int4_sim), (3) *attention destruction* (H2O-style). `exactkv_failures=0` across all 1,103 cells.

(1) Near-tie noise (int8): 66% of divergences are near-ties (margin < 0.05); lossy mean rank 2.4; late divergence (fdi=22). Small 8-bit noise flips only closely-contested decisions, explaining why int8 diverges on LongBench (uncertain outputs) but not structured tasks.

(2) Distribution shift (int4_sim): 83% top-1 flip, 26% near-tie, mean margin 0.305. The full-KV model is moderately confident, yet int4_sim selects ∼3rd–4th ranked token (mean rank 3.5), diverging at median token 8. Compression biases the activation toward plausible but incorrect alternatives.

(3) Attention destruction (H2O-style): 100% top-1 flip, 0% near-tie, mean margin 0.54, median fdi=1. Eviction eliminates context anchors from the very first generated token. The lossy model selects tokens ranked 6th–7th in the full-KV distribution.

These three modes directly explain the task-type hierarchy: near-tie noise only surfaces on long-context uncertain tasks (LongBench); distribution shift affects all tasks but worsens with longer generation; attention destruction is immediate regardless of task. `exactkv_failures=0` across all 1,103 divergent cells.

11 Downstream Task-Impact Metrics

Token-level drift is ExactKV's primary metric, but downstream task output validity is the ultimate concern. We report tool-call validity (BFCL), Python syntax (MBPP), and LongBench answer overlap from existing panel data.

Compressor	Model	n	Drift	Full-KV valid	ExactKV valid	Preserved
noop	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
noop	Mistral	200	0.0%	43/200 (21.5%)	43/200 (21.5%)	**100%**
int8	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
int8	Mistral	200	1.5%	43/200 (21.5%)	43/200 (21.5%)	**100%**
int4_sim	Llama	200	45.0%	63/200 (31.5%)	63/200 (31.5%)	**100%**
int4_sim	Mistral	200	55.5%	43/200 (21.5%)	43/200 (21.5%)	**100%**

Table 17: BFCL tool-call validity preservation (v2.7, 1,200 cells). Despite 45–55% token-level drift for `int4_sim`, ExactKV preserves **100%** of full-KV valid tool calls for both models. *Note: "preservation" means ExactKV matches full-KV output, not that full-KV itself always produces valid calls. The full-KV baseline achieves only 26.5% (106/400) valid tool calls on this panel.*

Category	n	Drift	Full-KV valid	ExactKV valid	Preserved
simple	104	49.0%	38/104 (37%)	38/104 (37%)	**100%**
parallel	104	54.8%	31/104 (30%)	31/104 (30%)	**100%**
multi_turn	104	59.6%	24/104 (23%)	24/104 (23%)	**100%**
ast_eval	88	35.2%	13/88 (15%)	13/88 (15%)	**100%**

Table 18: BFCL downstream validity by task category (`int4_sim`, 400 cells). ExactKV achieves 100% preservation across all four categories despite 35–60% drift.

Task	int4_sim div	int4_sim acc	int8 div	int8 acc
trec	83.3%	0.904	0.0%	1.000
multifieldqa_en	66.7%	0.955	16.7%	0.997
lcc (code)	66.7%	0.908	4.2%	0.994
qasper	100.0%	0.899	16.7%	0.994
samsum	91.7%	0.854	16.7%	0.993
gov_report	95.8%	0.781	16.7%	0.997
hotpotqa	100.0%	0.787	37.5%	0.972
2wikimqa	100.0%	0.880	29.2%	0.987
narrativeqa	100.0%	0.716	66.7%	0.973
passage_retrieval	100.0%	0.746	41.7%	0.974

Table 19: LongBench per-token acceptance rate as draft-utility proxy. Even when all cells diverge (100%), `int4_sim` achieves 72–95% per-token acceptance: ExactKV speculatively uses the lossy draft for most generation before correcting the divergent suffix. `int8` reaches 97–100% on the same tasks.

Compressor	n	F1 (full)	F1 (lossy)	F1 (ExactKV)	ExactKV=full
noop	240	0.043	0.043	0.043	**100%**
int8	240	0.043	0.042	0.043	**100%**
int4_sim	240	0.043	0.044	0.043	**100%**

Table 20: LongBench answer-overlap diagnostic (HF LongBench v2.6, 720 cells). Max token-F1 vs reference answers; *not* official LongBench scores. Short `max_new_tokens` limits absolute F1; value is relative parity vs full-KV.

Compressor	n	Drift	Full-KV valid	ExactKV valid	Preserved
noop	24	0.0%	3/24 (12.5%)	3/24 (12.5%)	**100%**
int8	24	0.0%	3/24 (12.5%)	3/24 (12.5%)	**100%**
int6_sim	24	0.0%	3/24 (12.5%)	3/24 (12.5%)	**100%**
int4_per_vec	24	0.0%	3/24 (12.5%)	3/24 (12.5%)	**100%**
int4_sim	24	12.5%	3/24 (12.5%)	3/24 (12.5%)	**100%**

Table 21: MBPP Python syntax validity (Llama v3.0, 96 cells). `ast.parse` on extracted code; *not* pass@1. Combined: 12/12 full-KV valid outputs preserved.

12 Scaling Analysis: Context-Length and Generation-Length

Threat / confound. Same as cross-panel: family rates mix task, context, and `max_new`. Below: within-task controls plus opportunity framing.

Compressor	2K context	4K context	8K context
noop	0.0%	0.0%	0.0%
int8	21.2%	27.5%	25.0%
int4_sim	**95.0%**	**86.2%**	**90.0%**

Table 22: Context-length scaling on HF LongBench v2.6 (80 cells each; fixed reading family). `int4_sim` is near-ceiling at all lengths — context is a weak within-family axis. `int8` is moderate (21–28%) and roughly flat.

Compressor	mnt=16	mnt=32	mnt=128	mnt=256
noop	0.0%	0.0%	0.0%	0.0%
int8	0.0%	0.0%	0.5%	1.0%
int4_sim	9.0%	13.5%	**38.5%**	**62.0%**

Table 23: Generation-length scaling on BFCL (both models; fixed tool-call family). `int4_sim` scales 7× from mnt=16 to mnt=256 — strongest within-task control. `int8` remains near-zero.

Family	Div. rate	Mean FDI	FDI/`mnt`	Hazard proxy
MBPP	6.2%	17.0	0.65	0.0027
BFCL short	11.2%	10.2	0.40	0.0051
BFCL long	50.2%	84.5	0.41	0.0039
LongBench	90.4%	7.8	0.17	0.080

Table 24: `int4_sim` length-opportunity metrics from committed panels (`reports/public_release/length_opportunity.json`). Hazard proxy = #divergent / $\sum$(tokens until first divergence or end); diagnostic only. BFCL long diverges late (FDI≈85); LongBench flips early (FDI≈8).

Combined: matched hierarchy on both models (Mistral 0%→97%, Llama 17%→91% at shared budgets); controlled BFCL length (9%→62%); weak controlled LongBench context (near-ceiling).

Family	Mistral	Llama
MBPP (code)	**0.0%** (0/54)	**16.7%** (9/54)
BFCL (tool)	**23.3%** (21/90)	**25.6%** (23/90)
LongBench (reading)	**96.7%** (87/90)	**91.1%** (82/90)

Table 25: Matched task×context×`max_new` factorial (`int4_sim`; shared 2K/4K/8K and mnt 32/64/128; 1,404 cells). Family hierarchy survives budget matching on both models. Prompts still differ by family. Artifacts: `reports/external_panels/matched_factorial/`.

No identical-prompt factorial. Budgets are matched; template text is not. `exactkv_failures=0` remains a harness invariant, not the headline.

13 int6_sim: Non-Catastrophic Intermediate Compressor

`int6_sim` implements 6-bit symmetric quantisation simulation (64 discrete levels, scale = max($|x|$)/31). It fills the quantisation-error gap between `int8` and `int4_sim`. GPU-validated (v3.0, both models): 0% BFCL/MBPP; 37.5%/47.2% LongBench (Mistral/Llama). Mean absolute error is 4.15× `int8` and 0.23× `int4_sim`.

Compressor	bits	levels	ratio vs fp16	quant. error	type
int8	8	256	0.50×	1.0× (baseline)	real
`int6_sim`	**6**	**64**	**0.375×**	**4.15×** int8	sim
int4_sim	4	16	0.25×	18.3× int8	sim

Table 26: `int6_sim` compression curve. GPU-validated (v3.0, both models): 0% BFCL/MBPP. 37.5% (Mistral) / 47.2% (Llama) LongBench. `exactkv_failures=0`.

14 int4_per_vec_sim: Per-Vector INT4 Quantisation

`int4_per_vec_sim` implements per-vector symmetric INT4 quantisation, inspired by the per-vector/per-channel granularity used in KVQuant/KIVI-style KV quantisation [2, 6]. Instead of a single scale over the entire KV tensor, each token-position vector of size `head_dim` receives its own scale: scale = max($|x|$)/7 computed independently for each `[batch, head, seq_pos]` vector. This eliminates cross-head quantisation contamination from outlier attention heads.

Quantisation error comparison on a realistic KV cache shape [1, 32, 512, 64] with simulated outlier heads (first 4 heads 5× larger activations):

Compressor	Granularity	MAE	×int8	Notes
int8	per-tensor	0.067	1.00×	Real compressor
int4_per_vec_sim	**per-vector**	**0.138**	**2.06×**	**KVQuant/KIVI-style**
int6_sim	per-tensor	0.275	4.10×	Intermediate
int4_sim	per-tensor	0.844	12.59×	Per-tensor catastrophic

Table 27: Per-vector INT4 achieves 6× lower error than per-tensor INT4 on realistic KV shapes with outlier heads (2.06× int8 vs 12.59× for per-tensor). Granularity dominates bit-width: `int4_per_vec_sim` (4-bit) beats `int6_sim` (6-bit, per-tensor).

Predicted vs observed v3.0 divergence (both models, task-family means):

Task family	int8	int4pv pred.	int4pv obs.	int6 obs.	int4 obs.
MBPP code	0%	∼0–2%	**0%**	0%	6.3%
BFCL tool-call	0%	∼1–4%	**0%**	0%	52.5%
HF LongBench	18.1%	∼30–50%	**56.3%**	42.4%	85.4%

Table 28: Analytical prediction vs GPU-observed divergence for `int4_per_vec_sim`. v3.0 panel (both models, `exactkv_failures=0`). LongBench observed value is slightly above the predicted range but remains between `int8` and per-tensor `int4_sim`.

Key insight: The per-vector vs per-tensor distinction reveals that quantisation *granularity* is at least as important as bit-width for KV cache compression on structured tasks. At 8K context, bit-width still contributes meaningfully.

Compressor	MBPP	BFCL	LongBench	Class
int8	0%	0%	18.1%	8-bit quant
int6_sim	0%	0%	42.4%	6-bit quant
int4_per_vec	0%	0%	56.3%	4-bit per-vector
int4_sim	6.3%	52.5%	85.4%	4-bit per-tensor
h2o_sim_75	,	,	100%	Eviction (75% kept)

Table 29: Core benchmark curve (both-model means. v3.0 quant compressors. H2O from v2.8). `exactkv_failures=0` throughout.

14.1 Faithful external adapter appendix

Phase D3 wires **faithful external adapters** (upstream algorithms, not ExactKV tensor sims) on the same crash-test grid as built-in compressors. Results are **appendix evidence only**, tracked separately from the 8,132-cell headline total. Reproduction: `REPRODUCE.md`. Internal wave labels (1–3) in tables below map to artifact directories in Appendix E.

Three roles (not three successes): `int8` is the non-catastrophic **real** baseline; `snapkv_experimental` tests whether kvpress SnapKVPress runs end-to-end; `kivi_offline_r32` is an integration diagnostic (not production KIVI). Wave-2/3 add KnormPress and `turboquant_experimental` (offline NumPy adapter, not production TurboQuant serving).

Table 30: Faithful adapter panel summary (1,568 appendix cells, not in headline 8,132).

Wave	Models	Cells	Compressors	Status
1	Llama+Mistral	864	int8, SnapKV, KIVI r32	Complete
2	Mistral only	128	int8, KnormPress, TurboQuant, SnapKV	Smoke
3	Llama+Mistral	576	int8, TurboQuant	Complete

Wave-1 (864 cells): `int8` is the only non-catastrophic real compressor (∼8–9% combined drift). SnapKV 90–97%. KIVI r32 100% (integration diagnostic).

Table 31: Wave-2 structured-task smoke, Mistral (128 cells). `exactkv_failures=0`.

Compressor	MBPP	BFCL	Combined
`int8`	0.0%	0.0%	0.0%
`turboquant_experimental`	6.2%	0.0%	3.1%
`kvpress_knorm_experimental`	75.0%	81.2%	78.1%
`snapkv_experimental`	87.5%	100.0%	93.8%

Wave-2 shows near-int8 TurboQuant drift on structured tasks only (Mistral, MBPP+BFCL smoke). It is not a faithful non-catastrophic baseline.

Table 32: Wave-3 full grid by task family (576 cells, both models). `exactkv_failures=0`.

Family	Cells	int8 div.	TurboQuant div.	TQ accept.
HF LongBench	288	16.7%	**64.6%**	0.868
BFCL	160	0.0%	5.0%	1.000
MBPP	128	0.0%	1.6%	0.998
Combined	**576**	**8.3%**	**34.0%**	**0.933**

Interpretation: `turboquant_experimental` is **task-conditional**: near-clean on structured code/tool tasks (1.6–5.0% drift on MBPP/BFCL) but ∼63–67% LongBench drift on both models. `int8` remains the only non-catastrophic real compressor in this appendix grid (8.3% combined).

Table 33: Wave-3 HF LongBench by model (144 cells each).

Model	**int8 div.**	**int8 accept.**	**TQ div.**	**TQ accept.**
Llama-3.1-8B	18.1%	0.992	62.5%	0.861
Mistral-7B	15.3%	0.985	66.7%	0.874

This reinforces the task-type hierarchy from built-in compressors; it does **not** establish a faithful non-catastrophic LongBench baseline. KIVI production CUDA remains blocked (Exp 024).

15 Forensic Divergence Case Studies

Three representative cells from the logit autopsy (Section 10) illustrate the mechanistic failure modes in detail.

15.1 Forensic Case A: int8: Near-tie noise

Rank	Token	Full-KV prob	Lossy prob	Note
1	`' very'`	0.129	0.113	full-KV argmax
2	`' fine'`	0.113	0.129	**int8 chose this**
3	`' pretty'`	0.083	0.082	
4	`' great'`	0.075	0.073	
5	`' good'`	0.065	0.064	

Table 34: int8, NarrativeQA, 4K context, token 53. Margin 0.016 (near-tie <0.05). INT8 noise flips `' very'` → `' fine'`, a near-synonym. `exactkv_failure=0`.

15.2 Forensic Case B: int4_sim: Distribution shift

Setting: NarrativeQA reading comprehension, 2K context, Llama-3.1-8B. The full-KV model is moderately confident (margin 0.158), yet `int4_sim` selects a rank-4 token. This is the distribution-shift signature: 4-bit quantisation biases attention activations so that activation energy accumulates on a plausible but incorrect continuation, even when the full-KV model's preference is clear. Divergence occurs at token 1 (the very first generated token), meaning the entire generation trajectory is corrupted from the start.

Rank	Token	Full-KV prob	Note
1	`','`	0.271	full-KV argmax
2	`' they'`	0.232	
3	`' and'`	0.202	
4	`' I'`	0.113	**int4_sim chose this (rank 4)**
5	`' it'`	0.039	

Table 35: int4_sim, NarrativeQA, 2K context, token 1. Margin 0.158 (full-KV confident). 4-bit quantization shifts attention activation toward rank-4 token. `exactkv_failure=0`.

15.3 Forensic Case C: H2O-style: Attention destruction

Setting: HotpotQA multi-hop reasoning, 2K context, Llama-3.1-8B, h2o_sim (50% tokens kept). This is qualitatively different from both Cases A and B: the H2O-style eviction discards the factual anchor tokens needed for multi-hop reasoning, leaving only attention-sink tokens and the recency window. The result is not a near-tie or a biased distribution, it is a *collapsed representation* where the evicted model selects a token that does not even appear in the full-KV top-5. The fact that the full-KV model distributes probability across numeric tokens (suggesting factual recall) while H2O selects a question mark (suggesting confusion) confirms total context destruction.

Rank	Token	Full-KV prob	Note
1	'1'	0.056	full-KV argmax
2	'15'	0.039	
3	'20'	0.035	
4	'10'	0.034	
5	'5'	0.034	
>5	'?'	<0.034	**H2O-style chose this (rank >5!)**

Table 36: h2o_sim (50% kept), HotpotQA, 2K context, token 1. H2O-style chose outside top-5 entirely, the evicted tokens included factual anchors. Collapsed representation. not a near-tie or distribution shift. `exactkv_failure=0`.

16 Divergence Case Studies (Release Panel)

Case	Compressor	1st	Acc.	Corr.	Lossy vs full snippet
A	int4_sim	3	0.75	1	art vs many
B	int4_sim	1	0.50	1	math drift at token 1
C	int4_sim	5	0.70	1	JSON: temperature vs country
D	int4_sim	8	0.94	1	late divergence at mnt=16
G	int4_sim	2	0.88	n/a	long_ctx 1024 (Mistral), verifier exact
O	kivi_offline	0	0.000	n/a	LB NarrativeQA: -!!!!... vs prose
P	int4_sim	1	0.00	1	BFCL export-50: schema drift

Table 37: Historical release panel case studies. See §10 for full autopsy across 1,103 divergent cells.

17 Kernel Microbenchmark

Kernel microbenchmark only, *not* end-to-end inference speedups.
Source: Phase-F kernel microbenchmark (Appendix inventory).
Test configuration: shape $[1, 8, 512, 64]$, CUDA.

Table 38: Torch vs Triton kernel latency (ms). Ratio = torch / Triton.

Mode	torch (ms)	Triton (ms)	Ratio
int8	0.115	0.0706	1.63×
int4	0.3319	0.2162	1.54×
block_sparse	0.7622	0.7797	0.98×*

*block_sparse uses the torch execution backend only (not Triton-accelerated).

17.1 Verifier overhead cost model (qualitative + proxy)

Strict systems reviewers ask how verification cost scales as draft acceptance drops. ExactKV does **not** yet publish per-token verifier latency on the 1,500-cell headline panel. We provide a simple amortized model plus an evidence-plus wall-clock proxy.
Amortized model (serving intuition): let α be mean draft acceptance. Expected verify work grows as α falls: E[verify rounds] $\approx (T_{\text{gen}}/L) \cdot (1 - \alpha)$ for draft length L. Lower acceptance implies more full-KV forward passes amortized over the generation.

Table 39: Evidence-plus wall-clock proxy (144 cells). Harness runs full + lossy + ExactKV sequentially; cell time is flat across compressors. *Not* per-token serving latency.

Compressor	Mean acceptance	Divergence	Mean cell (s)
noop	1.000	0.0%	3.84
int8	1.000	0.0%	3.87
int4_sim	0.985	31.3%	3.85

On LongBench (Table 19), `int4_sim` acceptance drops to 0.72–0.95 where divergence is 100%, implying substantial verify amortization under the model above even though ExactKV measures rather than optimizes that cost. Artifact: systems verifier-overhead summary (Appendix inventory).

17.2 Systems diagnostic panel (peak CUDA + path wall-clock)

Practical memory/latency questions are answered by a separate **96-cell** `systems_diagnostic` panel (both models; shared 2K/4K × mnt 64/128; `noop`/`int8`/`int4_sim`). Each cell records per-arm `gpu_peak_allocated_bytes` (process-level peak CUDA allocation) and `wall_clock_ms` (harness path timing) for full / lossy / ExactKV. Hardware: NVIDIA RTX PRO 4000 Blackwell, torch 2.8.0+cu128, float16; `exactkv_failures=0` on all 96 cells.

Claim boundary: diagnostic peak CUDA and harness wall-clock — *not* serving RPS, TTFT, or unqualified production VRAM savings. Peak includes model weights + KV + temporaries. ExactKV peaks higher than lossy-only (full + compressed state coexist) and is ∼2.3× slower than full/lossy on this harness (verify cost). Runner: `scripts/run_systems_diagnostic_panel.sh`; pack: `reports/systems/systems_diagnostic.json`.

Table 40: Systems diagnostic peak CUDA allocation (GiB, mean; n=16 cells/arm).

Model	Comp.	full	lossy	ExactKV
Llama-3.1-8B	noop	16.10	16.49	16.49
Llama-3.1-8B	int8	16.10	16.67	16.72
Llama-3.1-8B	int4_sim	16.10	16.67	16.72
Mistral-7B	noop	14.23	14.67	14.68
Mistral-7B	int8	14.23	15.22	15.26
Mistral-7B	int4_sim	14.23	15.22	15.26

Table 41: Systems diagnostic path wall-clock (ms, mean; harness timing, not serving TTFT/RPS).

Model	Comp.	full	lossy	ExactKV
Llama-3.1-8B	noop	3690	3746	8472
Llama-3.1-8B	int8	3691	3713	8689
Llama-3.1-8B	int4_sim	3660	3675	8572
Mistral-7B	noop	3566	3620	8275
Mistral-7B	int8	3571	3585	8389
Mistral-7B	int4_sim	3535	3141	8345

17.3 Serving microbench (TTFT-like + serial requests/sec)

A follow-up **76-cell** `serving_microbench` panel (72 strong + 4 `serial_16` extras) records TTFT-like latency, completed-requests/sec under **serial** multi-request load, and peak CUDA / Δ vs full for full / lossy / ExactKV on the same HF harness (not continuous batching, not vLLM). Hardware:

NVIDIA RTX PRO 4000 Blackwell; `exactkv_failures=0` on all 76 cells. Strong design: both models × {`noop`,`int8`,`int4_sim`} × ctx {2048,4096} × mnt {64,128} × n_{req} {1,4,8}.
Claim boundary: HF multi-request diagnostic — *not* vLLM integration, continuous batching, production serving, or unqualified VRAM savings. Across strong-panel loads, ExactKV completes ∼1.9× fewer requests/sec than full (∼0.13 vs ∼0.25) and shows ∼1.5× higher TTFT-like latency (∼940 ms vs ∼630 ms). Runner: `scripts/run_serving_microbench_panel.sh`; pack: `reports/systems/serving_microbench.json`.

Table 42: Serving microbench completed requests/sec (mean over serial_1/4/8).

Model	Comp.	full	lossy	ExactKV
Llama-3.1-8B	noop	0.243	0.247	0.132
Llama-3.1-8B	int8	0.247	0.245	0.129
Llama-3.1-8B	int4_sim	0.247	0.246	0.129
Mistral-7B	noop	0.248	0.253	0.136
Mistral-7B	int8	0.253	0.250	0.134
Mistral-7B	int4_sim	0.253	0.371	0.132

Table 43: Serving microbench TTFT-like latency (ms, mean over serial_1/4/8).

Model	Comp.	full	lossy	ExactKV
Llama-3.1-8B	noop	646	611	939
Llama-3.1-8B	int8	610	623	943
Llama-3.1-8B	int4_sim	611	617	943
Mistral-7B	noop	666	626	946
Mistral-7B	int8	625	637	947
Mistral-7B	int4_sim	625	631	947

18 VeriCache: Closest Prior Art

VeriCache [8] is closer than a casual skim suggests. Both use compressed draft + full-KV verify/correct and target lossless greedy equivalence. **ExactKV must not claim** to invent that loop, VeriCache owns it for **lossless serving**.

Table 44: VeriCache vs ExactKV, complementary goals (ExactKV does *not* reproduce VeriCache).

	VeriCache	**ExactKV**
Goal	Serve faster, same outputs	Measure drift / first divergence
System	GPU compressed KV, offloaded full KV, staggering	Evaluation harness only
Metrics	Throughput, tiering	`first_divergence`, acceptance, failure
Reproduced?	n/a	**No**

19 Why ExactKV Still Matters After VeriCache

VeriCache uses verification to serve faster without changing outputs. ExactKV uses verification to measure exactly where and how compressors drift.

1. Compressor-agnostic crash-test on a fixed prompt panel.
2. Lossy-path first divergence (unverified `generate_lossy_greedy`).

3. Public diagnostic leaderboard with frozen cell definitions.
4. Explicit taxonomy: lossy drift vs acceptance vs `exactkv_failure`.

ExactKV does *not* replace VeriCache for deployment.

20 Related Work

Verification-based serving. VeriCache [8] is the closest prior art. Both systems use a compressed-KV draft path and a full-KV verify/correct loop. The distinction is purpose: VeriCache is a throughput-oriented *serving system* (scheduling, memory tiering, cross-resource staggering). ExactKV is a *diagnostic measurement framework* reporting where the unverified compressed path first diverges and how often. ExactKV does not reproduce VeriCache's system design and makes no claim to the draft+verify algorithm. Speculative decoding [3] and MagicDec [1] use draft/verify for throughput acceleration. ExactKV borrows the same semantics purely for measurement.

KV quantization methods. KVQuant [2] proposes non-uniform per-channel INT4 quantization targeting KV outlier heads. KIVI [6] proposes asymmetric 2-bit per-channel KV quantization with a CUDA kernel. Both directly motivate `int4_per_vec_sim`, which simulates per-vector granularity without a production kernel. The `kivi_offline` adapter (real KIVI quantizer math, no CUDA/Triton kernels) revealed 100% divergence in the current offline integration (Table 11), a diagnostic, not a claim about KIVI's algorithmic accuracy. SnapKV [4] selects important KV entries by eviction. The `h2o_sim` family models the H2O Heavy Hitter Oracle policy [9] from the same compressor class and shows 100% LongBench divergence even at 75% retention (Section 8).

KV storage and streaming. CacheGen [5] compresses KV caches for network streaming and bandwidth reduction. LMCache [7] targets KV reuse and offloading across serving instances. Neither is designed to measure when or why the compressed path first diverges from the full-KV oracle.

Evaluation methodology. Standard benchmarks (LongBench, BFCL, HumanEval) measure task-level accuracy averaged over many tokens. ExactKV's first-divergence index (fdi) is a token-resolution diagnostic orthogonal to task accuracy: a cell can score perfectly on a task metric while accumulating 90% token-level drift (if the verifier corrects it). ExactKV uses official HF LongBench and BFCL datasets but reports *drift measurements*, not official benchmark scores.

21 Novelty and Claim Boundaries

Defensible (narrow): compressor-agnostic diagnostic/leaderboard for token-level drift and first-divergence under verifier semantics.

Not novel: compressed draft + full-KV verify (VeriCache), draft-verify for acceleration (speculative decoding), lossy-KV divergence observation (VeriCache problem statement).

ExactKV is *not* a production serving system and does *not* reproduce VeriCache. Compression ratios are stored tensor byte ratios.

22 Future Work and Limitations

Completed: 144-cell evidence-plus (512/1024). 216-cell Llama-only external smoke (LongBench/RULER/BFCL/HumanEval bundled pilots). 144-cell MBPP smoke (both models). **640-cell `kivi_offline` panel** over real HF LongBench + MBPP subsets (see Table 11). `kivi_offline` uses real KIVI quantizer math (`models.utils_quant` simulate path. no CUDA/Triton kernels). The panel reveals catastrophic adapter-level drift (100% divergence, acceptance ≈ 0) and all cells corrected (`exactkv_failures=0`).

Future work, KIVI production path: The faithful adapter appendix is **complete** (1,568 cells; Section 14.1). **Remaining:** production KIVI CUDA/Triton kernel validation (Exp 024 blocked), not adapter smoke. The legacy `kivi_offline` result (no residual) is an adapter diagnostic.

Completed extended panels: BFCL tool-call validity (1,200 cells, mnt=128/256, both models, `exactkv_failures=0`); H2O-style token-eviction (800 cells, five variants, both models); top-k logit autopsy across 1,103 divergent cells (`--store-top-k-logits`), revealing three mechanistically distinct failure modes. Total: **8,132 completed GPU cells**.

Remaining future work: expand BFCL validity to larger/more diverse prompts; MBPP pass@1 with safe test execution; RULER 12K+; HELMET; InfiniteBench 100K+; per-token verifier latency on headline panels; broader scaling curves (12K+, 16K+ context, more compressors); production KIVI/KVQuant/SnapKV CUDA kernels (adapter smoke complete; Exp 024 blocked).

Unvalidated proxy slots. Raw headline artifacts may contain placeholder compressor slots that are *not* real external integrations; they are excluded from all ranked tables and scientific claims. Headline evidence uses `noop`, `int8`, `int4_sim`, and the separately labeled simulated / faithful families only.

Matched factorial (both models complete). Shared context 2K/4K/8K and `max_new` 32/64/128 across MBPP/BFCL/LongBench yields `int4_sim` Mistral **0%/23%/97%**, Llama **17%/26%/91%** (Table 25). The older cross-family confound is addressed for budgets; prompts still differ by family.

Systems / serving diagnostics are not production serving. Peak CUDA and path wall-clock (`systems_diagnostic`, 96 cells) and TTFT-like / completed-requests/sec (`serving_microbench`, 76 cells) are HF harness diagnostics — not vLLM RPS, continuous batching, or unqualified VRAM savings. ExactKV is $\sim$1.9$\times$ lower completed-req/s and $\sim$1.5$\times$ higher TTFT-like under serial load.

23 Conclusion

VeriCache shows how to **serve** with compressed KV without changing outputs. ExactKV shows **where and why compressors drift** on the lossy path, what the unguarded path would ship, and how often verification corrects.

ExactKV's strongest supported claim is that **KV-cache drift is governed jointly by task type, generation length, compressor class, and quantization granularity**, with generation length as the cleanest within-task control, cross-family spans as the observational hook, and a matched-budget factorial confirming the hierarchy on both models. Mapped with mechanistic autopsy, unguarded lossy damage, and a three-family downstream validity story.

Across **8,132 completed GPU cells**, ExactKV establishes four empirical findings:

1. **Task-family hierarchy survives matched budgets.** Matched factorial: `int4_sim` Mistral 0%/23%/97%, Llama 17%/26%/91% (MBPP/BFCL/LongBench) at shared context and `max_new`. The older 6%→90% cross-panel span remains a useful hook.

2. **Generation length is the strongest within-task driver.** On BFCL, `int4_sim` scales from 9% (mnt=16) to 62% (mnt=256), a 7× increase. LongBench flips early (FDI≈8); BFCL long accumulates late (FDI≈85).

3. **Compressor class determines failure mode.** Logit autopsy over 1,103 divergent cells reveals: (a) near-tie noise (int8, mean rank 2.4, fdi=22); (b) distribution shift (int4_sim, rank 3.5, fdi=8); and (c) attention destruction (H2O-style, rank 6.7, fdi=1).

4. **Unguarded lossy damage; verification recovers full-KV-valid outputs.** Without verify, wrong tool/code fields can ship. With ExactKV: BFCL 318/318, MBPP 12/12 (Llama), LongBench overlap 100% vs full-KV. `exactkv_failures=0` is the harness invariant, not the headline.

Two new compressors extend the compression curve: `int6_sim` (6-bit per-tensor) and `int4_per_vec_sim` (4-bit per-vector, KIVI/KVQuant-style). An extended validation panel (both models, 1,568 cells, `exactkv_failures=0`) validates both as non-catastrophic: 0% on BFCL/MBPP; 37–47% (int6) and 56–57% (int4 per-vector) on LongBench. Per-vector granularity eliminates drift on structured tasks but 4-bit resolution still accumulates at 8K context.

Faithful upstream adapters (1,568 appendix cells): a full `int8` vs `turboquant_experimental` grid on both models shows TurboQuant is near-clean on structured tasks but ∼63–67% LongBench drift; `int8` remains the only non-catastrophic real compressor in that grid. This reinforces task-type sensitivity; it does not establish a faithful non-catastrophic LongBench baseline.

ExactKV's contribution is not a new compressor. It is a measurement infrastructure that makes KV-cache drift **observable, attributable, and reproducible**, and a verifier mechanism that corrects it with zero exactness failures across all 8,132 tested cells.

A All Completed Panels: Benchmark Card

Panel	Models	Ctx (K)	Cells	EKV fail
Headline release	Llama+Mistral	0.5–2	1,500	0
Evidence-plus	Llama+Mistral	0.5,1	144	0
Ext. smoke: LongBench	Llama only	2,4	72	0
Ext. smoke: RULER 2K–4K	Llama only	2,4	48	0
Ext. smoke: RULER 8K	Llama only	8	24	0
Ext. smoke: BFCL pilot	Llama only	1,2	48	0
Ext. smoke: HumanEval	Llama only	1,2	24	0
MBPP code-drift smoke	Llama+Mistral	0.5,1	144	0
BFCL export-50 tool-call	Llama+Mistral	1,2	1,200	0
KIVI offline adapter	Llama only	2,4	640	0
HF LongBench v2.6	Llama+Mistral	2,4,8	720	0
BFCL validity v2.7	Llama+Mistral	1,2	1,200	0
H2O token-eviction v2.8	Llama+Mistral	2,4	800	0
v3.0 int6/int4pv (Mistral)	Mistral	2,4,8+	784	0
v3.0 int6/int4pv (Llama)	Llama	2,4,8+	784	0
Subtotal (pre-v3.0)			6,564	
Subtotal (v3.0)			1,568	
Grand total (headline)			**8,132**	**0**
Faithful wave-1	Llama+Mistral	2,4,8+	864	0
Faithful wave-2 smoke	Mistral	0.5,1	128	0
Faithful wave-3	Llama+Mistral	2,4,8+	576	0
Faithful appendix total			**1,568**	**0**
Matched factorial	Llama+Mistral	2,4,8	1,404	0

Table 45: All completed ExactKV GPU panels as of v3.0. `exactkv_failures=0` throughout. Reconciliation: $6{,}564 + 784 + 784 = 8{,}132$ headline cells. Faithful appendix (1,568 cells) and matched factorial (1,404 cells) tracked separately. Each v3.0 row = one model × four compressors × three task families (784 cells). Not official benchmark scores.

B Divergence Autopsy: Extended Details

See Section 10 for the full mechanistic analysis and logit autopsy table. This appendix section provides the methodology reference: at each divergence position, ExactKV records the top-5 full-KV token probabilities and the actual lossy-chosen token. Metrics: top-1 flip (different argmax), near-tie (full-KV margin < 0.05), mean lossy rank (where the lossy-chosen token ranks under full-KV distribution), and median first-divergence index (fdi). Script: `scripts/analyze_logit_margins.py`. Panels with stored top-k logits: v2.6 LongBench, v2.7 BFCL validity, v2.8 H2O-style.

C Reproducibility

All quantitative claims are backed by committed JSON artifacts under `reports/` (Appendix inventory). Readers do **not** need to open the repository to follow the scientific argument; the repo is for verification and re-execution.

Public artifact: git tag `v-release` · https://github.com/utkarshg20/ExactKV

Goal	How
CPU verify (∼2 min, no GPU)	`REPRODUCE.md` · `bash scripts/smoke_test.sh`
Check committed reports	`python3 scripts/exactkv_repro.py --reports-only`
Rebuild this PDF	`bash scripts/build_paper_pdf.sh` (tectonic)
GPU re-run of cited panels	Full command inventory in `REPRODUCE.md`

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